Smart Parking System

Mandatory Constraints:

Budget:5000 USD

Existing CCTV cameras

No Budget for New Sensors

SMART objectives

1. Reduce employee parking search time to under 2 minutes.
2. Enable pre-booking for at least 90% of parking sessions.
3. Achieve 95% successful automated digital payment transactions.
4. Ensure parking slot detection accuracy of at least 85% using cameras and Cloud Vision API.
5. Provide daily admin monitoring and reporting of bookings, payments, and occupancy.

Updated Scope Features

1. Parking Slot Pre-Booking
   - Employees can reserve a parking slot in advance using a web or mobile app.
   - Ensures a slot is available upon arrival.

2. Camera-Based Slot Detection
   - Uses existing CCTV cameras to detect whether a parking spot is occupied or vacant**.**
   - Vehicle detection is handled using Google Cloud Vision API trial**.**

3. Basic Barrier Gate
   - A manual or simple automated barrier gate is used at the entry/exit.
   - Gate operation may involve manual confirmation by security or a basic trigger (e.g., RFID/QR code scan).
   - Provides controlled access without requiring high-cost automation**.**

4. App Navigation to Assigned Slot
   - App guides the employee directly to their reserved parking spot, minimizing search time.

5. Automated Digital Payment
   - Parking fees are processed digitally during booking or after parking.
   - Payment gateways like Stripe, Razorpay, or PayPal are integrated.

6. Booking & Payment Confirmation Notifications
   - Employees receive **r**eal-time notifications confirming slot assignment and successful payment.

7. Admin Dashboard & Monitoring
   - Admin can monitor occupancy, bookings, and payments**.**
   - Provides basic reporting for operational oversight.

8. Basic Data Security & Access Control
   - Role-based access ensures only authorized users can book slots or access admin functions.

Expected Outcomes:

Reduced Time Spent on Parking

- Employees can **pre-book a parking slot** before arrival
- Existing CCTV cameras detect slot occupancy
- Employees are guided directly to the allocated spot

Outcome:
Parking search time reduced from 10–15 minutes to under 1 minute

Efficient Use of Existing Infrastructure

- Uses existing parking spots and CCTV cameras
- No need for installing sensors in every slot

Outcome:
Reduce Budget

Pre-Booking Experience

- Employees book parking slots via web/mobile app
- Slot availability updated using camera-based detection

Outcome:
Reduces the Time Spent on parking

Improved Employee Productivity & Satisfaction

- Reduced stress and delays
- Predictable and reliable parking experience

Outcome:
Higher employee satisfaction and recovered work hours

Automated & Transparent Payment

- Parking fee collected digitally during booking or exit
- Automatic invoice generation

Outcome:
No cash handling, no billing disputes

Potential Failure:

1. Camera Detection Accuracy Risk

Risk: Poor lighting, shadows, or camera angles may lead to **incorrect detection of occupied/vacant parking slots**.

Mitigation:

- Adjust and optimize camera angles and placement.
- Periodically verify slot detection manually.
- Apply image pre-processing (contrast/brightness adjustments) before sending to Cloud Vision API.

2. Basic Barrier Gate Failure

Risk: Barrier gate may malfunction or fail to open/close properly.

Mitigation:

- Schedule **regular maintenance** of the gate.
- Provide **manual override** for security staff in case of gate failure.
- Keep **backup access cards/QR codes** for entry.

3. Network or Cloud Service Downtime

Risk: Internet outage or Cloud Vision API downtime may **disrupt slot detection or pre-booking updates**.

Mitigation:

- Implement **local caching** of last-known slot status.
- Retry API requests automatically when network is restored.
- Allow **manual entry and booking override** during downtime.

4. Payment Gateway Failure

Risk: Payment transactions may fail due to gateway downtime or errors.

Mitigation:

- Enable **pay-at-exit** or offline fallback option.
- Automatically retry failed transactions.
- Notify employees immediately of payment issues.

5. Data Privacy & Security Risk

Risk: Unauthorized access to employee data, payment info, or CCTV images.

Mitigation:

- Encrypt sensitive data in transit and at rest.
- Implement **role-based access control**.
- Avoid storing raw camera images unnecessarily; use only processed results.

Overall Role of APIs in the System

API Type	Role in Project
Vision API	Vehicle detection
Booking API	Slot allocation
Payment API	Cashless transactions
Auth API	Secure access
Notification API	User communication

2-Week Planning Phase – Tasks and Timeline

Task	Duration	Key Milestone
1. Requirement Gathering	2 days	Finalized requirements
2. Feasibility Study (Hardware, Cloud, API)	2 days	Feasibility approved
3. System Architecture Design	2 days	Architecture diagram ready
4. Component Selection (Existing Cameras, Server, APIs)	2 days	Hardware/Software list finalized
5. Cloud & API Evaluation (Google Cloud Vision API trial setup)	2 days	API setup complete

6. Cost Estimation & Budget Allocation	1 day	Budget approved
7. Risk Assessment & Mitigation Planning	1 day	Risk plan ready
8. Project Timeline & Milestone Definition	1 day	Project schedule finalized
9. Planning Review & Approval	1 day	Planning phase complete

Budget Requirement

1. Hardware Costs

Item	***Quantity***	***Unit Cost (USD)***	***Total (USD)***	***Notes***
Wi-Fi Router / LAN setup	1	100	100	Ensures connectivity for cloud and backend
Local mini-server / PC	1	300	300	Optional for local caching and backend processing
RFID Reader + 50 Tags	1 set	200	200	Optional automated entry/exit
Mounting & Power Accessories	—	—	100	Cables, adapters, mounts

2. Software & Development Costs

Item	***Cost (USD)***	***Notes***
Backend Development	600	Handles booking, slot status, API integration
Frontend Development	600	Employee web app + admin dashboard
Cloud Vision API Integration	400	Connects existing cameras to Google Cloud Vision API trial for vehicle detection
Payment Gateway Integration	300	Stripe/PayPal/Razorpay integration for automated payments
Testing & Deployment	300	Unit testing, integration testing, deployment setup

3. Cloud & API Costs

Item	***Cost (USD)***	***Notes***
Google Cloud	0	Uses free trial credits (sufficient for pilot/initial release)

Vision API Trial		
Cloud Hosting / Database	150	AWS/GCP small instance or Firebase/ Supabase (free tier for trial, small paid instance for reliability)
Domain + SSL	30	For web app access and secure HTTPS

4. Labor Costs (Project Team)

Role	Cost (USD)	Notes
Software Developer (Backend)	600	2–3 weeks of work for initial MVP
Frontend Developer	600	Web app / dashboard setup
Integration & Testing	400	API integration, payment testing, camera calibration

5.Contingency / Buffer

Item	Cost (USD)	Notes
Contingency / Buffer	500	For unexpected expenses, additional sensors, extra API calls, or minor hardware upgrades

6.Total Budget Allocation

Category	Cost (USD)
Hardware	700
Software & Development	2,200
Cloud & APIs	180
Labor	1,600
Contingency / Buffer	500
TOTAL	**$5,000**